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Transparent Graphene/BN-Graphene Stacked Nanofilms for Electrocatalytic Oxygen Evolution

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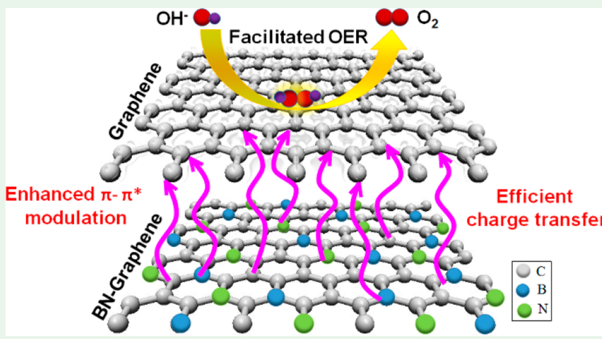
ABSTRACT: The fabrication of a carbon (C)-based, metal-free, efficient, and stable electrocatalyst at high oxygen evolution reaction (OER) overpotential is essential to advance the field of water splitting and air-battery related energy systems. For this purpose, we here synthesize a stacked nanofilm (G-BNG) of pure graphene (G) on boron and nitrogen (BN)-codoped graphene (BNG) with a total thickness of about 2−3 nm and 87% transparency at 550 nm that shows an efficient OER performance, better than that of a stacked G-G layer, G-BG film, or G-NG film in an alkaline electrolyte. The G-BNG film performs OER with the highest reaction kinetics (lowest Tafel slope ~ 143.22 mV/dec) and a small overpotential (0.58 V) along with 81.6% retention of the current density at 1.8 V (vs reversible hydrogen electrode) even after 50000 s of operation. Experimental analysis reveals that the doped B and N atoms in the bottom layer influence the π-bonding environment in the top G layer, facilitating the effective electron/charge transfer for OER electrocatalysis. Furthermore, the demonstrated OER performance is comparable or even superior to that of other reported C-based electrocatalysts with at least 400−700 times lower catalyst loading, making the G-BNG film promising for advanced renewable energy conversion systems, such as various kinds of transparent air batteries and water electrolyzers.



KEYWORDS: *BN-doped graphene, stacked nanofilm, C-based transparent electrocatalysts, oxygen evolution reaction, interlayer charge transfer*

1. INTRODUCTION

Modern technological advancements on portable electronics systems have led to a seminal growth in global energy consumption.[1] To pave the way for innovative and uninterrupted technological advancement, it is essential now more than ever to develop clean, innovative, and renewable energy sources, such as fuel cell, metal−air battery, and water electrolyzer.[2−5] The conventional forms of these systems have limitations related to either their high cost due to the use of precious metal catalysts or impediments for a lack of understanding as well as fabrication complexity. Therefore, intense research works are being performed to develop highly efficient, metal-free, and environmentally friendly energy conversion and storage devices.[6−8] The efficiency of these renewable energy systems mainly depends on the performance of electrode materials associated with different reactions, such as cathodic oxygen reduction reaction (ORR), anodic oxygen evolution reaction (OER), and hydrogen evolution reaction (HER).[9−11] Efficient OER is particularly essential for advanced air batteries and water-splitting systems.[9] Thermodynamically, OER is an uphill reaction involving a stepwise four-electron transfer at a high overpotential. However, the high overpotentials and corresponding slow kinetics of OER hinder the energy efficiency greatly.[10,11] The most efficient OER electrocatalysts are still based on noble-metal/metal-oxide, such as Pt, IrO_2, and RuO_2.[12,13] However, high cost and inadequacies impede the widespread utilization of these noble-metal-based as well as non-noble-metal-based other electrocatalysts, which appeared as alternatives.[11] As such, because of the interfacial complexity of the hybrid structures in non-noble-metal catalysts and the related thermodynamic instability, they often suffer from low selectivity, poor durability, and/or susceptibility to gas poisoning with adverse environmental effects.[11] Consequently, as a low-cost and metal-free alternative, carbon (C)-based metal-free nanomaterials have been introduced for efficient electrocatalytic applications.[10−16] For example, graphene, as a two-dimensional (2D) atomically thin network of sp^2-hybridized C atoms, has been examined extensively. Regrettably, graphene itself does not perform excellent OER activity compared to the commercially available costly metal-based catalysts because of a lack of active catalytic

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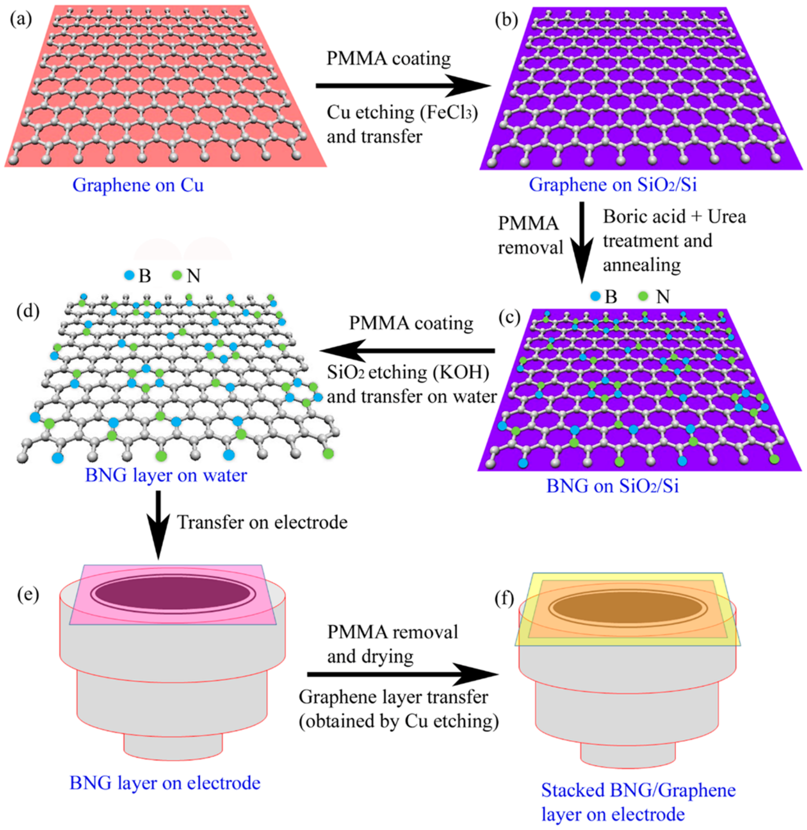

Figure 1. Schematic diagram of the synthesis procedure of BNG (a−d) and stacking of G-BNG layers onto a glassy-carbon electrode surface (e and f).

sites on the graphene lattice.[11,17] However, proper functionalization of graphene with other heteroatoms, such as N, S, B, and P, is an effective way of tuning their electronic properties in order to achieve enhanced catalytic properties.[18−21] Recently, atomic-substitution-based heteroatom codoping of graphene-based materials with B and N or N and P has demonstrated promising OER and HER activities.[20−22] Amperometric theoretical calculation has revealed that not only does the electronegativity difference between heteroatom(s) and a C atom dictate the electrocatalytic activity of heteroatom-doped graphene-based materials but also the doping-induced charge and/or spin redistribution(s) play(s) an important role in enhancement of the electrocatalytic activities in such catalysts.[9,23] Despite all of these efforts, the heteroatom-doped graphene is often observed to remain unstable under a high OER overpotential in O-containing electrolytes, which needs serious research attention.[11]

Moreover, researchers have recently been focusing on C-based metal-free three-dimensional (3D) porous electrocatalysts to achieve high OER efficiency, but the unified design motives and underlying catalytic mechanism remain somewhat unclear.[9,10,24−26] This is reflected in several recent research attempts in designing 2D-layered catalysts.[20,27,28] In one of such attempts, black P on N-doped graphene (NG) flakes exhibited efficient water splitting.[20] It has also been demonstrated that exfoliated single-layer nanosheets prepared from NiFe- and NiCo-based layered double hydroxides exhibit higher catalytic activity than their 3D forms and outperformed a state-of-the-art IrO_2 catalyst.[27] In another study, researchers reported that the interfacial stress generated by the stacking of 2D-heterostructured graphene and MoS_2 nanosheets facilitated Li-ion insertion through an energy-efficient cation-exchange transformation.[28] Undoubtedly, serious research attention must be provided to reinvestigate the 2D material-based metal-free electrocatalyst development in more scientifically fundamental ways. Here we demonstrate, for the first time, that even a 2D graphene-based ultrathin catalyst can be an efficient OER electrocatalyst if the reaction mechanism and electrokinetics are well interpreted. In this attempt, we have stacked pure graphene onto heteroatom (B or N)-doped (BG or NG) or codoped (B and N) graphene (BNG) as an efficient OER catalyst. We have adapted a simple chemical treatment of few-layer graphene for uniform B or N doping as well as BN-codoping, although B doping of sp^2-hybridized C is very difficult compared to N doping even through other complex techniques.[29,30]

The heteroatom-doped graphene often loses its fascinating conductive properties due to doping-induced electron scattering, which, in turn, hinders its electrocatalytic activity by worsening carrier transport in the planar direction. However, the stacking of graphene and doped graphene could preserve the following purposes: (i) efficient vertical charge transfer onto the pure graphene layer from the bottom heteroatom-doped or codoped graphene occurs; (ii) the top graphene layer overcomes direct doping-induced conductivity degradation; (iii) stacking-induced $\pi-\pi^*$ orbital modulation improves the electrochemical OER activity; (iv) high OER stability at 1.8 V versus reversible hydrogen electrode (RHE) is achieved. Moreover, the stacked-double-layer (SDL) catalyst is (i) 87% transparent, (ii) only 2−3 nm in thickness, and (iii) comparable or superior in OER performance in an alkaline medium with at least 400−700 times lower catalyst loading

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($\sim$0.68 μg/cm^2) on the electrode in comparison to other reported C-based catalysts with heteroatoms or metal doping in 2D and even 3D architectures. The excellent OER catalytic stability of G-BNG is very promising to achieve stable and reliable OER electrocatalysts. We have further investigated the underlying mechanism of the OER activity of the stacked catalyst layer and correlated the structural and chemical properties with the OER performance.

2. EXPERIMENTAL METHODS

Materials Synthesis. Few-layer graphene (G) was grown on Cu foil (25 μm thick) using thermal chemical vapor deposition (T-CVD). The Cu foil was first placed inside a quartz tube and reduced by heating at 1000 °C with 65 sccm hydrogen (H_2) and 200 sccm argon (Ar) gas flow for 20 min to remove the oxide layer from Cu surface. Thereafter, 10 sccm methane (CH_4) was added to the H_2 and Ar gas flow for 10 min to grow graphene. The as-grown graphene was then transferred onto a SiO_2-coated Si substrate after etching the Cu substrate with a 0.1 M $FeCl_3$ solution in deionized (DI) water. During the transfer, the graphene layer was protected by a poly(methyl methacrylate) (PMMA; 4% in anisole) coating, which was removed from the graphene surface by keeping it under hot acetone (65 °C) for 30−40 min. The graphene surface was then treated in mild O_2 plasma ($\sim$20 W) for 30 s to make it hydrophilic. For the synthesis of N-doped graphene (NG), the hydrophilic graphene film on the SiO_2/Si substrate was immersed in 1 M urea (CON_2H_4) solution in DI water inside a closed vial and heated at 100 °C for 15 min, whereas the B-doped graphene (BG) was made using 0.1 M boric acid (H_3BO_3) solution under the same conditions. Similarly, the BNG film was synthesized by using a mixed solution of 0.1 M boric acid and 1 M urea taken at 1:1 ratio under the same conditions. The chemically treated graphene samples were finally heated at 900 °C in a quartz tube for 1 h in 100 sccm Ar gas flow separately. Afterward, the NG, BG, or BNG layer was coated with PMMA, the SiO_2 layer was etched out in a 2 M KOH solution kept at 60 °C (overnight), and the layer was transferred onto the electrode or a substrate after float-washing it onto a DI water surface several times. The SDLs were mounted by first transferring the heteroatom-doped graphene layer onto glassy-carbon electrode surface, followed by drying at 60 °C for 30 min and PMMA removal in hot acetone (65 °C) for 30 min (three times). Thereafter, the pure graphene layer was transferred onto the heteroatom-doped graphene layer surface and dried at 60 °C. Finally, the PMMA layer from the graphene surface was removed by heating the sample in hot acetone again at 65 °C for 30 min (three times). A schematic diagram of the detailed experimental procedure for stacked-layer fabrication onto a glassy-carbon electrode has been presented in Figure 1. For a simple demonstration, we used a single layer of graphene in the schematic diagram. Four different types of SDLs were fabricated, namely, G-G, G-NG, G-BG, and G-BNG, either on the electrode surface or on other substrates (such as, glass or Si) for electrochemical testing or different characterizations, respectively.

Materials Characterization. For transmission electron microscopy (TEM) imaging, a small piece (5 mm^2) of the stacked and dried double-layer G-BNG film on the SiO_2/Si substrate was coated with PMMA, the SiO_2 layer was etched away in a 2 M KOH solution at 60 °C, and the double-layer film was washed by floating onto fresh DI water several times. Finally, the double-layer film was cut through the middle part for TEM imaging. The film was then transferred onto a C-coated Cu grid, and the PMMA layer was removed by using hot acetone, as mentioned earlier. TEM images were recorded by FEI Tecnai TF20 field-emission gun. Scanning electron microscopy (SEM) images were taken using a high-resolution field-emission scanning electron microscope (JEOL JSM-6510LV/LGS at 10 kV). The chemical composition of the films was characterized by X-ray photoelectron spectroscopy (XPS) on a VG Microtech ESCA 2000 spectrometer using a monochromic Al X-ray source (97.9 W and 93.9 eV). XPS data analysis was done with the commercially available *CasaXPS* software (www.casaxps.com), and the peaks were fitted to a Gaussian and/or Lorentzian function if not Gaussian−Lorentzian to get a better fitting and acceptable full-width at half-maximum. The charge correction of the spectra was done by placing the main C 1s component at 284.5 eV as a reference for graphitic C.[31] Elemental mapping was obtained using energy-dispersive X-ray spectroscopy (EDS) under SEM. The Raman spectra were collected by Raman spectroscope (Renishaw) using a 514 nm green laser with a 50× air objective and a laser power of 20 mW for an accumulation time of 15 s and averaging over three scans. For recording optical absorbance and transmittance spectra, different films were fabricated on glass substrates and a UV−vis spectrometer was used.

Electrochemical Testing. Electrochemical measurements were performed on an electrochemical workstation (CHI760C, CH Instrument, U.S.A.) with a three-electrode electrochemical cell. All experiments were conducted at room temperature (25 °C). Typically, a Pt wire was used as the counter electrode, a saturated calomel electrode was mounted as a reference electrode in an O_2-saturated 0.1 M KOH electrolyte, and the potentials were converted to RHE. The SDLs were fabricated on the glassy-carbon electrode as described above.

3. RESULTS AND DISCUSSION

Figure 2a presents the TEM image of the G-BNG double-layer film, while the inset shows only the BNG film with 1−2 layers, as has been pointed out with arrows. It clearly depicts the layer thickness of the G-BNG double layer. The graphene film consists of $\sim$5 graphene monolayers, while the BNG film contains 2 monolayers. A long-range TEM image has been shown in Figure S1a,b. The decrease in the layer thickness of the BNG film may be attributed to the O_2 plasma treatment as

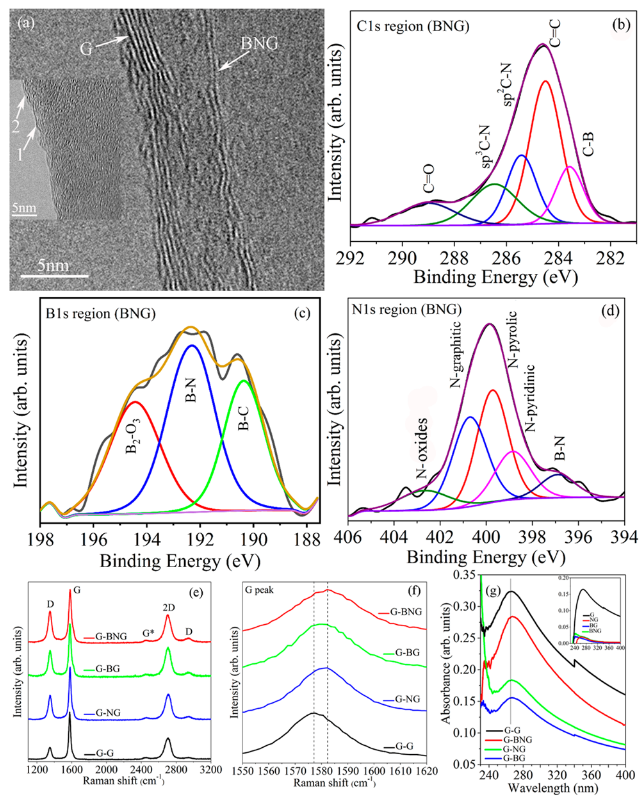

Figure 2. (a) TEM image of the G-BNG double-layer film surface. The inset shows only the BNG film. Representative high-resolution XPS spectra: (b) C 1s (c) B 1s, and (d) N 1s regions for the BNG film. Raman spectra of different double-layer films: (e) full range; (f) G peak positions. (g) Optical absorbance spectra for double-layer films. The inset shows the absorbance spectra for different films.

well as chemical treatment, followed by annealing at 900 °C. Long-range TEM images for the G-BNG film, the optical images of various double-layer films, and SEM images of different films are presented in Figures S1−S3, respectively. The observed gap between the two layers (Figure S1) under the TEM images is not real as at the edge the two layers might not have similar lengths on the TEM grid and such difference in edge lengths appears as a gap. The average double-layer nanofilm thickness is about 2−3 nm.

For chemical composition analysis, representative high-resolution XPS spectra for the C 1s, B 1s, and N 1s regions of the BNG film are shown in parts b−d of Figure 2, respectively, while the full range XPS spectra and other high-resolution spectra for different films are presented in Figure S4. The C 1s, N 1s, and B 1s spectra for the NG, BG, and BNG films are located respectively around the 284, 400, and 190 eV binding energies. The N content in the NG film is 8.46 atom %, while the BG film contains 12.14 atom % B. The BNG film contains B and N contents of about 10.89 and 9.63 atom %, respectively. The detailed chemical compositions are evaluated by deconvolution of the C 1s, N 1s, and B 1s spectra. The C 1s peak positioned at 284.5 eV for all of the films arises mainly from graphitic C═C bonding between sp^2-hybridized C atoms.[31] The deconvoluted C 1s and N 1s spectra for the NG film are presented in parts c and d of Figure S4, respectively. The subpeaks at 285.9 and 287.1 eV under C 1s can be attributed to N atoms bonded to sp^2- and sp^3-hybridized C atoms, respectively.[32] Their contents are 3.1 and 5.4 atom %, respectively, as given in Table S1. The subpeak around 288.7 eV is related to O atoms bonded to C atoms (content of ∼10.3 atom %). On the other hand, the N 1s spectrum (Figure S4d) can be attributed to four different types of bonding in a graphene lattice, namely, N pyridinic (at 398.6 eV and 3.3 atom %), N pyrolic (at 399.5 eV and 2.7 atom %), N graphitic (at 400.6 eV and 1.8 atom %), and N oxides (at 402.1 eV and 0.6 atom %).[33,34] For BNG film (parts b−d of Figure 2) under the C 1s spectrum, C−B, C═C, C−N, and C−O groups are observed. The C−B bond is located at 283.5 eV (11.03 atom %), while the C═C, sp^2-hybridized C−N, and sp^3-hybridized C−N peaks are spotted at 284.5 eV (31.86 atom %), 285.4 eV (14.48 atom %), and 286.7 eV (13.57 atom %), respectively.[31,32,34] The B 1s peak is constituted with B−N, B−C, and B_2O_3 peaks. The N 1s peak can be subdivided into five peaks related to B−N (at 397 eV and 0.77 atom %), N pyridinic (at 398.8 eV and 1.66 atom %), N pyrolic (at 399.7 eV and 3.36 atom %), N graphitic (at 400.7 eV and 2.91 atom %), and N oxides (at 402.6 eV and 0.94 atom %) respectively.[34,35] Detailed chemical contents for the NG, BG, and BNG films are given in Table S1. From XPS analysis, N atoms in the graphene lattice for the NG film is mostly pyridinic and pyrolic in nature, while the B and N codoping changes the N-bonding environment in the graphene lattice, making mostly N pyrolic species. For the BNG film, the decrease in the N graphitic content can be attributed to B−N bond formation. The N pyridinic content also increases in the BNG film compared to that in the NG film. EDS elemental mapping of B, C, and N in the BNG film was also performed, and the spectra are presented in Figure S5, which shows the uniform distribution of B and N heteroatoms into the graphene lattice. Thus, N, B, and BN codoping in graphene were achieved by a simple chemical technique.

Raman spectra of different SDL films are collectively shown in Figure 2e. The main Raman peaks D, G, and 2D are observed for all of the films.[36−38] The D peak originates from zone-boundary phonon scattering induced by sp^3 defect sites when the graphene lattice is subjected to light,[36] while the G peak appears because of planar sp^2 C═C stretching vibrations in a crystalline graphitic material[37,38] and the 2D peak represents a second-order resonance of the D peak. The D, G, and 2D bands for the double-layer films are located near 1351, 1580, and 2705 cm^{-1}, respectively. Detailed observation reveals that the G peak position for the G-G film is located at 1577 cm^{-1}, which blue-shifts to 1581.7 cm^{-1} for G-NG, 1579.8 cm^{-1} for G-BG, and 1582.6 cm^{-1} for G-BNG (Figure 2f). This shift may be attributed to the stacking of two layers as well as different heteroatom/heteroatoms doping in the bottom layers. To investigate this shifting effect, Raman measurements on only the G, NG, BG, and BNG layers are also performed, and the spectra are presented in Figure S6a. The G peak positions for the G film is 1576.6 cm^{-1}, which is very close to that in the double-layer G-G film. The slight increase (about 0.4 cm^{-1}) in the G peak position in the G-G film, as compared to that of only the G film, may be correlated to the trapped charges or air molecules within the double-layer interface or the stacking-induced structural distortions across the interface. However, the G peak positions for the NG, BG, and BNG films are at 1581.4, 1580.5, and 1582.8 cm^{-1}, respectively (Figure S6b). Xue et al. have also reported such a blue shift of the G peak upon heteroatom doping into the graphene lattice.[39] Such shifting can be attributed to the structural distortion of graphene associated with the different bond lengths of C−C, C−N, and C−B induced by doping as well as the generated strain into the graphene lattice.[39−43] Therefore, the observed blue shifts in the G peak positions for the NG, BG and BNG films are induced by heteroatom doping into the graphene lattice, whereas the G peak shifts for the G-NG, G-BG, and G-BNG double-layer films are influenced mainly by heteroatom/heteroatoms doping into the bottom graphene layers and stacking-induced charge transfer onto the graphene layer and consequent strain within the SDL films. The small peaks on both sides of the 2D peak can be attributed to the G* and D + G bands.[44]

The intensity ratio of the D and G Raman peaks (I_D/I_G ratio) reveals defect or defect-induced charge density in C-based materials.[45] The I_D/I_G ratio for the G-G double layer is 0.26, which increases to 0.47 and 0.5 for the G-NG and G-BG double layers. This ratio is maximum (0.6) for the G-BNG double-layer film, which reflects the presence of the highest defect density (Figure S6c). The I_D/I_G ratios for the G, NG, BG, and BNG layers were also estimated and are presented in Figure S6d. The I_D/I_G ratio for the G film (0.26) is similar to that for the G-G double layer. However, for the NG, BG, and BNG films, this value is increased to 0.84, 0.82, and 0.97, respectively, which has also been reported by other researchers.[39,46,47] This observation implies that the heteroatoms doped in the graphene lattice induces doping related to charge transfer in the double-layer films, which behave as defects under Raman measurement. Such charge transfer can induce localized strains in the graphene lattice, resulting in an increased I_D/I_G ratio for hybrid double-layer films. Another important structural feature for the Raman spectrum for C materials is the appearance of a G peak shoulder, marked as the D′ band, which is also associated with the lattice defect density. The D′ band positions for the G, NG, BG and BNG films are located at 1609, 1616, 1617, and 1621 cm^{-1}, respectively (Figure S6a), which may be due to the structural difference of

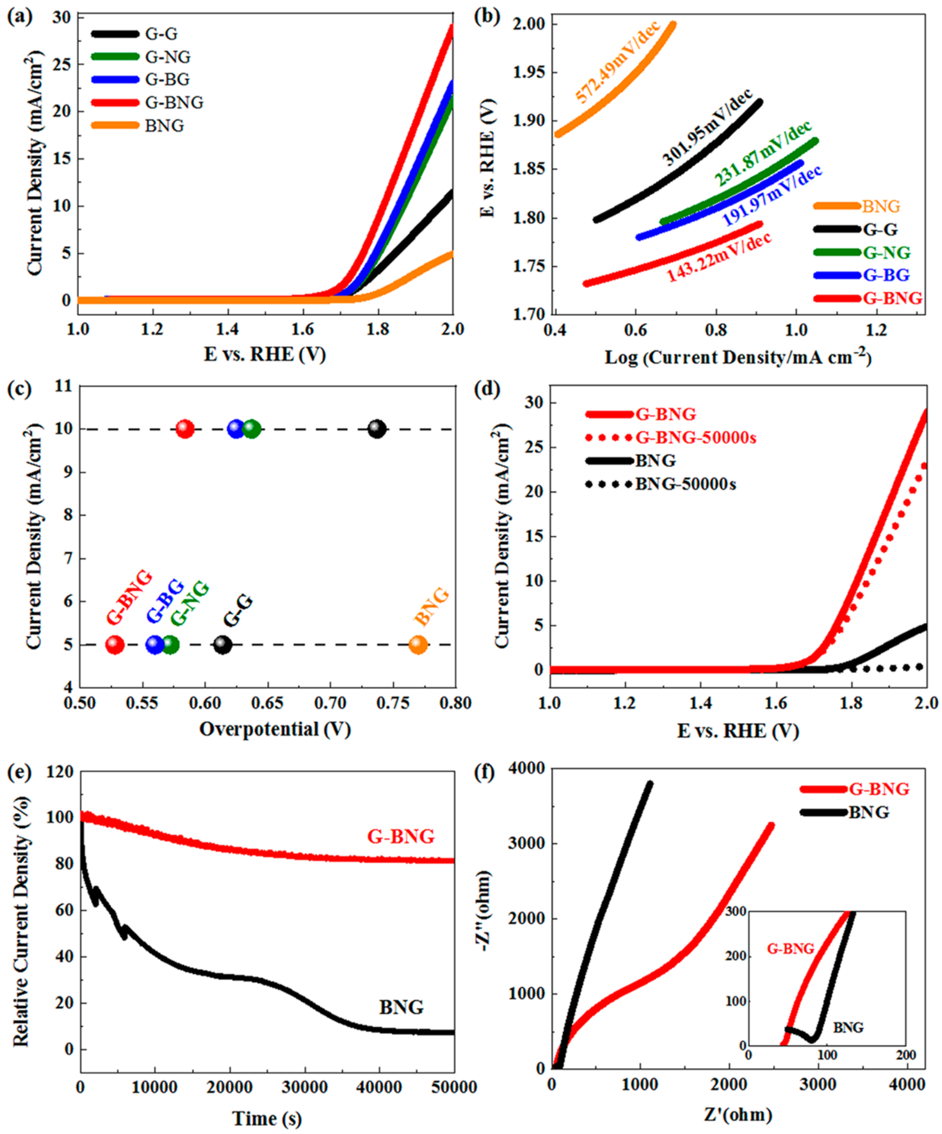

Figure 3. Electrocatalytic activities toward OER in a 0.1 M O_2-saturated KOH solution at room temperature: (a) OER polarization curves of G-BNG, G-NG, G-BG, G-G, and BNG at a scan rate of 10 mV/s; (b) Tafel plots of G-BNG, G-NG, G-BG, G-G, and BNG derived from part a; (c) overpotential of G-BNG, G-NG, G-BG, G-G, and BNG to achieve current densities of 5 and 10 mA/cm; (d) OER polarization curves of G-BNG and BNG before and after long-term stability measurements (50000 s) at a scan rate of 10 mV/s; (e) *I*−*t* curves of G-BNG and BNG at a constant potential of 1.8 V (vs RHE); (f) impendence spectra of G-BNG and BNG. Inset: enlarged EIS curves of G-BNG and BNG.

the defect types through different heteroatom(s) doping into the graphene lattice.[39,47] Furthermore, the intensity of this peak is highest for the BNG film. A similar D′ band is also observable in double-layer films having a maximum intensity for the G-BNG double-layer film, indicating higher defect density in the BNG layer and charge-transfer-induced strain in the G-BNG double-layer film. Therefore, the stacking of heteroatom-doped and pure graphene layers induces charged transfer onto the top graphene layer, which can, in turn, modify the electronic properties of the graphene layer and may promote efficient electrochemical activities.

To further enlighten the impact of the mechanism for charge transfer in the SDL film, optical absorbance measurements were performed on different films. Figure 2g shows the absorbance spectra of different SDL films, and in the inset, the absorbance spectra of one-layered films are shown. The optical absorbance is highest in the G film (~0.17) compared to doped films, which is attributed to the excitation of π bonds in the graphene lattice, and this can be related to the abundant $\pi-\pi^*$ transitions upon interaction with light in the UV−vis range with a maximum intensity at 269 nm (~ 4.61 eV). Recently, Imamura and Saiki also reported the interaction of UV light with $\pi-\pi$ stacking in transferred multilayer graphene.[48] The absorbance intensity of doped graphene films is much lower. The inset of Figure 2g shows the comparative absorbance spectra of NG and BG with absorbance maxima at 244 nm (~5.08 eV) and 264 nm (~4.69 eV), respectively, with an absorbance intensity of 0.02 for both films. For the BNG film, there are two distinct absorbance shoulders related to N and B doping, and the positions of these shoulders (~5.08 and 4.69 eV) exactly match those of the NG and BG films (Figure S7). Therefore, heteroatom doping into the graphene lattice influences the π-bonding environment and, hence, the $\pi-\pi^*$ transition. This phenomenon can be explained by an interband excitonic transition in the graphene film.[49−51] In an exciton, the electron

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and hole may have either parallel or antiparallel spins. The spins are coupled by the exchange interactions, which are generally influenced by heteroatom doping in the graphene lattice, as observed for heteroatom-doped graphene films.

The absorbance spectra of the G-G double-layer film have a highest intensity of 0.33 with a maximum absorbance at 266 nm (~4.66 eV), whereas the absorbance intensities of the G-NG and G-BG double-layer films are 0.19 and 0.17, respectively. Interestingly, their absorbance maxima are around 266.6 nm (~4.65 eV) and 267 nm (4.64 eV), respectively, which reflects that the excitonic transition is greatly modified in hybrid SDL films. The G-BNG double-layer film shows an absorbance intensity of 0.3, which is close to that of the G-G double-layer film (0.33) and has a maximum intensity at 268 nm (~4.62 eV), which is the lowest among double-layer films. Therefore, stacking of the BNG and pure graphene films has greatly influenced the π-bonding environment by increasing the probability of a $\pi-\pi^*$ excitonic transition through charge transfer, which could influence the catalytic property of the G-BNG double-layer film. According to the photonic band gap estimated using Tauc's equation, the band structure was also found to be modified through heteroatom doping and stacking (Figure S8). The excitonic transition in the G-G film did not change compared to that in the G film (Table S2). However, heteroatom doping or codoping and stacking clearly influence the excitonic transitions, which may modulate their electrochemical properties. We have also recorded the transmittance spectra of different films, and the spectra are presented in Figure S9. The transmittance of NG, BG, and BNG layers is around 97−98%, which implies that these films possess of 1−2 layers (Figure S9a), whereas the 88−89% transmittance of the G film reveals its layer thickness to be 4−5 monolayers. These observations are consistent to the findings in TEM images because each monolayer graphene reduces about 2.3% optical transmittance.[49,50] The transmittance of different SDL films (Figure S9b) is 87−88% for hybrid double layers, which is quite transparent.

Figure 3 shows the electrocatalytic OER activities for G-G, G-NG, G-BG, and G-BNG double-layer films measured in O_2 -saturated 0.1 M KOH at room temperature (25 °C). Figure 3a presents the polarization curves for different double-layer films, and the related onset potentials for OER on the BNG, G-G, G-NG, G-BG, and G-BNG films are 1.77, 1.70, 1.67, 1.66, and 1.60 V versus RHE. The comparison of the OER performance among the BG, NG, and BNG films is presented in Figure S10a, whereas the reproducibility of the excellent OER performance is demonstrated in Figure S10b for three G-BNG samples. The Tafel plots obtained from polarization curves are shown in Figure 3b. A lower Tafel slope (143.22 mV/dec) on the G-BNG double-layer film compared to those of the G-BG (191.97 mV/dec), G-NG (321.87 mV/dec), G-G (301.95 mV/dec), and BNG (572.49 mV/dec) films is observed, indicating more favorable kinetics toward OER on the G-BNG film. We further compared the overpotentials required to deliver 5 and 10 mA/cm^2 current densities in Figure 3c. As can be seen, the overpotential for the G-BNG film is only 0.58 V, which is the lowest compared to those of other double-layer and BNG films. The first principle calculation performed by the Xia group has demonstrated that, in codoped graphene structures, the p-electron clouds from two heteroatoms overlap and interact with each other, generating more active sites on neighboring C atoms compared with single-element-doped graphene.[52] The observation of higher catalytic activity as well as lower overpotential in the G-BNG film can be related to the doping- and stacking-induced charge transfer and facilitated $\pi-\pi^*$ excitonic transitions. B−N codoping into the bottom graphene layer facilitates charge transfer onto the top graphene layer, which modifies the catalytic interaction and electron-transport property of the graphene layer required for the OER performance. On the other hand, modification in the π-bonding environment (lower band gap for the G-BNG film; Table S2) through charge transfer in the G-BNG double-layer film is influenced by the lower OER overpotential.

To examine the effect of the stacking of double layers on the electrochemical stability, we presented OER polarization curves of G-BNG and only BNG films before and after long-term (50000 s) stability at 10 mV/s scan rate in Figure 3d. After 50000 s of the G-BNG double layer, the OER catalytic activity is far better compared to that of only the BNG layer. Therefore, in the case of the G-BNG double layer, the OER activity is stable compared to that of the BNG film. Therefore, direct interaction of the BNG film with electrolyte for OER is unfavorable for long-term catalytic activity. Commensurate with this observation for the screening effect of π bonds toward higher catalytic activity, Imamura et al. also reported that the damage of the graphene layer by UV light is most unlikely when $\pi-\pi$ stacking is available, as in the case of a multilayer graphene.[48] Figure 3e quantitatively demonstrates that, after 50000 s of OER activity, the BNG layer retains only 7.7% relative initial current density, which may be due to the OER etching effect at high overpotential, whereas G-BNG holds almost 81.6% of it at a constant potential of 1.8 V (vs RHE). Nyquist plots of the electrochemical impedance spectroscopy (EIS) spectra of the BNG and stacked G-BNG films are shown in Figure 3f for comparison. The spectra imply that the overall resistance of the G-BNG layer is higher in the electrolyte compared to only the BNG layer.[53] This can be related to the higher layer thickness of the G-BNG film compared to only the BNG film, as observed by the TEM image. However, the enlarged EIS spectra, as shown in the inset, indicate that the electron-transfer resistance for the BNG layer is higher compared to that for the G-BNG double layer, which may be attributed to B−N doping into the graphene lattice, resulting in an increase of resistance. Upon stacking of the BNG film with a pure graphene film, electronic transfer at the electrode−electrolyte interface is facilitated through charge transfer and modulation in the electronic structure through an increase in the $\pi-\pi^*$ excitonic transition, which, in turn, increases the OER performance of the SDL films. However, ion diffusion through a stacked layer is hindered in comparison to only the BNG film, which is obvious due to the higher thickness of the G-BNG film. Therefore, for such a 2D-layered transparent electrocatalyst, charge transfer is more important than ion transfer toward a better OER performance because the electronic charge-transfer resistance is lower compared to the ionic transfer for the stacked G-BNG film. The EIS spectra recorded before and after the OER stability test for G-BNG does not change much, as shown in Figure S10c, demonstrating again its excellent OER stability. This observation brings forward the advantage of stacking of heteroatom-codoped and undoped graphene films.

It is previously observed that the sp^2-hybridized C materials (graphene, nanotubes, etc.) are rich in abundant π electrons, and these π electrons, after proper activation, take crucial role for electrocatalysis in the ORR.[54,55] The popular technique for this activation is to provide more electrons in the C π system through the doping of sp^2-hybridized C materials with electron-rich N atoms.[51] N atoms in the sp^2-hybridized C lattice impart extra electrons, which increases the electron density, raises the highest occupied electronic orbital, and facilitates the ORR. Furthermore, the delocalized π electrons can also be activated by conjugation with vacant orbitals, as in the case of electron-deficient B incorporation into a sp^2-hybridized C lattice, which also increases the catalytic activity.[29] In the case of BNG, the compensation effect by electron-efficient and -deficient dopants may lead to totally different electronic structures, with different conjugation effects in the delocalized π system.[42,56] The π electrons can remain unactivated or activated. Qiao et al. incorporated N and B sequentially into the graphene domain and demonstrated an enhanced synergistic coupling effect, which can facilitate the electrocatalytic activity toward the ORR.[56] In our study, the stacked G-NG and G-BG films show better OER activities compared to only the NG and BG films, which may be attributed to the coupling of activated π electrons of the bottom doped layer with the abundant π electrons on the graphene layer through charge sharing between two stacked layers across the interface. This coupling may lead to induced π-electron activation, which is reflected through OER catalytic enhancement. In the case of the G-BNG double-layer film, this induced activation effect of π electrons is more prominent because of heteroatom codoping, which furthermore, enhances the related OER activity. Figure 4 shows the schematic mechanism for OER electrocatalysis in the G-BNG stacked nanofilm. As such, Geim et al. reported that sandwich heterostructures with van der Waals or covalent interactions might have more flexibility in modifying their catalytic activity because of modulation in the surface work function and electronic states of the whole stack rather than only those on the topmost layer.[57] Such an effect can be attributed to the electric field formed at the interface of two layers separating electrons and holes, and the local strain due to stacking-induced surface reconstruction.[57,58] As such, the work function of graphene or reduced graphene oxide was also reported to be modified by bringing them in close proximity or in contact with MoS_2.[59,60] Charge transfer between the two layers creates a dipole moment across the atomic planes,[58] which has been demonstrated here for fine control of the chemical reactivity on the surface. Furthermore, the stacked G-BNG film demonstrated comparable or even superior OER activities, with at least 400−700 times less catalyst loading (~0.68 $\mu g/cm^2$) on the electrode than other reported C-based electrocatalysts (~0.3−0.5 mg/cm^2) with heteroatoms or metal doping and codoping in 2D and even 3D architectures, as presented in Table S3. To further investigate the electrochemical activity of three stacked layers, we have also stacked three films with heteroatom-doped and undoped graphene films in various combinations. However, the stacking of more than two graphene layers increases the interfacial resistance considerably, and the electrocatalytic activities of the stacked layers degrade.

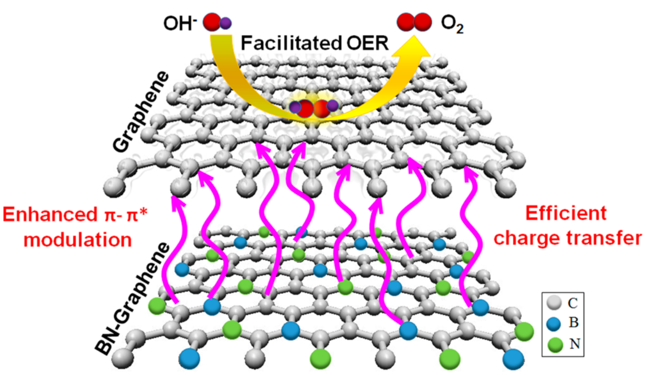

Figure 4. Electrocatalytic OER mechanism through efficient π-electron modulation and efficient charge transfer in a G/BNG stacked nanofilm.

4. CONCLUSIONS

We have doped CVD-grown G with B, N, and BN using a simple and scalable chemical functionalization technique, followed by annealing at 900 °C. TEM study reveals that the G film contains about 5 monolayers, whereas BNG possesses 1−2 monolayers. The doping amount is around 8−12 atomic % of B and N, as estimated by XPS analysis. Upon stacking of two hybrid layers, such as G-NG, G-BG, and G-BNG, for the first time, we have demonstrated that the electrocatalytic OER characteristics for the G-BNG film are greatly enhanced in comparison to those of all other films. The OER stability of the G-BNG layer is also observed to be excellent, with a retentivity of 81.6% after 50000 s at 1.8 V (vs RHE). As a whole, the G-BNG electrocatalyst is (i) 87% transparent, (ii) only about 2−3 nm thick, and (iii) comparable or superior in the OER performance in an alkaline medium with at least 400−700 times lower catalyst loading (~0.68 $\mu g/cm^2$) on the electrode as compared to other reported C-based catalysts (~0.3−0.5 mg/cm^2) with heteroatom(s) or metal doping in 2D or even 3D architectures. The results are attributed to the efficient interfacial charge transfer arising from heteroatom-codoping-induced π-electron modulation across the stacked layers and the corresponding local electrostatic strain due to the electronegativity difference between C, B, and N atoms. The excellent OER electrocatalytic performance along with the attractive OER stability of the transparent G-BNG nanofilm is promising for advancing not only the contemporary metal−air battery and water electrolyzer systems but also the smart, portable, transparent, and flexible energy devices in the near future.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsanm.0c02307.

TEM image of the G-BNG double layer, optical microscopy images of different SDLs (G-G, G-NG, G-BG, and G-BNG), SEM micrograph of the G, BNG, and G-BNG films, XPS spectra of different samples with detailed analysis, EDS elemental mapping of the BNG film, Raman spectra of the G, NG, BG, and BNG films with I_D/I_G ratios, Tauc's analysis of the band gaps from absorption spectra in the UV−vis wavelength range, optical transmission spectra, OER polarization curves of NG, BG, BNG, repeatability of OER for G-BNG, EIS of G-BNG before and after OER stability test, and a comparison of the OER performance with that of other reports (PDF)

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Author Contributions

R.P. and M.W. performed the fabrication, characterization of the samples, and data analysis. R.P. wrote the manuscript, and A.R. reviewed it. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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ABBREVIATIONS

CVD = chemical vapor deposition
OER = oxygen evolution reaction
PMMA = poly(methyl methacrylate)
RHE = reversible hydrogen electrode
SCCM = standard cubic centimeters per minute
SDL = stacked double layer
XPS = X-ray photoelectron spectroscopy

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